Paleomagnetism of parts of the Late Triassic diabase dike system associated with the trans-New Brunswick aeromagnetic lineament

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A paleomagnetic study was conducted on dike samples taken from three widely separated localities along the northeasterly trending Caraquet dike system (New Brunswick) in order to define their age relationship. This dike system is recognized on the basis of a well defined aeromagnetic lineament. Stable magnetization directions were isolated after alternating field (15–40 mT) and thermal (300–550°C) treatments of the samples taken from the three localities: C (353°, +56°), N (353°, +50°), and T (14°, +37°). The virtual pole derived from dike T (82°E, 62°N) agrees well with previously determined Late Triassic – Early Jurassic (200–190 Ma) poles for the Appalachians. On the other hand, the other two virtual pole positions from dikes C (143°E, 78°N) and N (133°E, 74°N) fall in the general area of average paleopoles for the Middle–Late Jurassic (180–170 Ma) age. The possibility that the dike at T differs in age from that of the dikes at localities C and N by at least 20 Ma is suggested. In the absence of reliable radiometric data, it is premature at the present stage to predict a likely occurrence of "younger" magnetization in these dikes related in origin to viscous partial thermoremanence possibly associated with the rifting and opening of the Atlantic in Middle Jurassic. Further investigations on newly found diabase outcrops along the magnetic lineament are in progress.

Une étude paléomagnétique a été réalisée sur des échantillons de dykes provenant de trois sites éloignés les uns des autres le long de la direction nord-est du système de dykes Caraquet (Nouveau-Brunswick) afin de préciser leur relation en âge. Ce système de dykes est identifié à partir d'un linéament aéromagnétique bien tracé. Des directions stables d'aimantation ont été isolées à la suite de traitements par champs artificiels (15–40 mT) et thermique (300–550°C) des échantillons prélevés aux trois sites: C (353°, +56°), N (353°, +50°) et T (14°, +37°). Le pôle virtuel dérivé du dyke T (82°E, 62°N) concorde bien avec les pôles préalablement attribués au Triassique supérieur – Jurassique inférieur (200–190 Ma) pour les Appalaches. D'autre part, les deux autres positions des pôles virtuels des dykes C (143°E, 78°N) et N (133°E, 74°N) se situent en âge autour de la moyenne des paléopôles du Jurassique moyen à supérieur (180–170 Ma). La possibilité que le dyke T diffère en âge des autres dykes aux sites C et N d'au moins 20 Ma est proposée. Sans données radiométriques fiables, il est prématuré au stade actuel de prédire une occurrence vraisemblable d'une aimantation "plus jeune" dans ces dykes dont l'origine serait reliée à une thermo-rémanence partielle visqueuse probablement associée à la fissuration et à l'ouverture de l'Atlantique au Jurassique moyen. D'autres recherches sont en cours sur de nouveaux affleurements découverts récemment le long du linéament magnétique.

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Introduction

In eastern North America, the Late Triassic was a period of normal faulting, graben development, and continental deposition of sediments. This was followed by extensive tholeiitic volcanism with the extrusion of basaltic lavas and dike injection as indicators of crustal rifting associated with the opening of the Atlantic Ocean (Ballard and Uchupi 1975; Uchupi and Austin 1979; Austin *et al.* 1980). Examples of Late Triassic igneous activity in the Canadian Appalachians (Fig. 1). include: (1) the North Mountain Basalt on the eastern shore of the Bay of Fundy; (2) basalts on Grand Manan Island and on islands in the upper reaches of the Bay of Fundy (Crosby 1962; Klein 1962; Jansa and Wade 1975; Wark and Clarke 1980); (3) basalts in offshore New Brunswick, recognized by their magnetic signature (Haworth and Keen 1979); (4) the Shelburne dike, exposed on shore for a distance of about 100 km along the southeastern coast of Nova Scotia (Taylor 1960) with its possible offshore extension (Haworth and Keen 1979); (5) a northeasterly striking, narrow, and discontinuous dike system that extends from Caraquet (northern New Brunswick) to Milo (Maine), a distance of about 500 km, recognized by Burke *et al.* (1973) from ground

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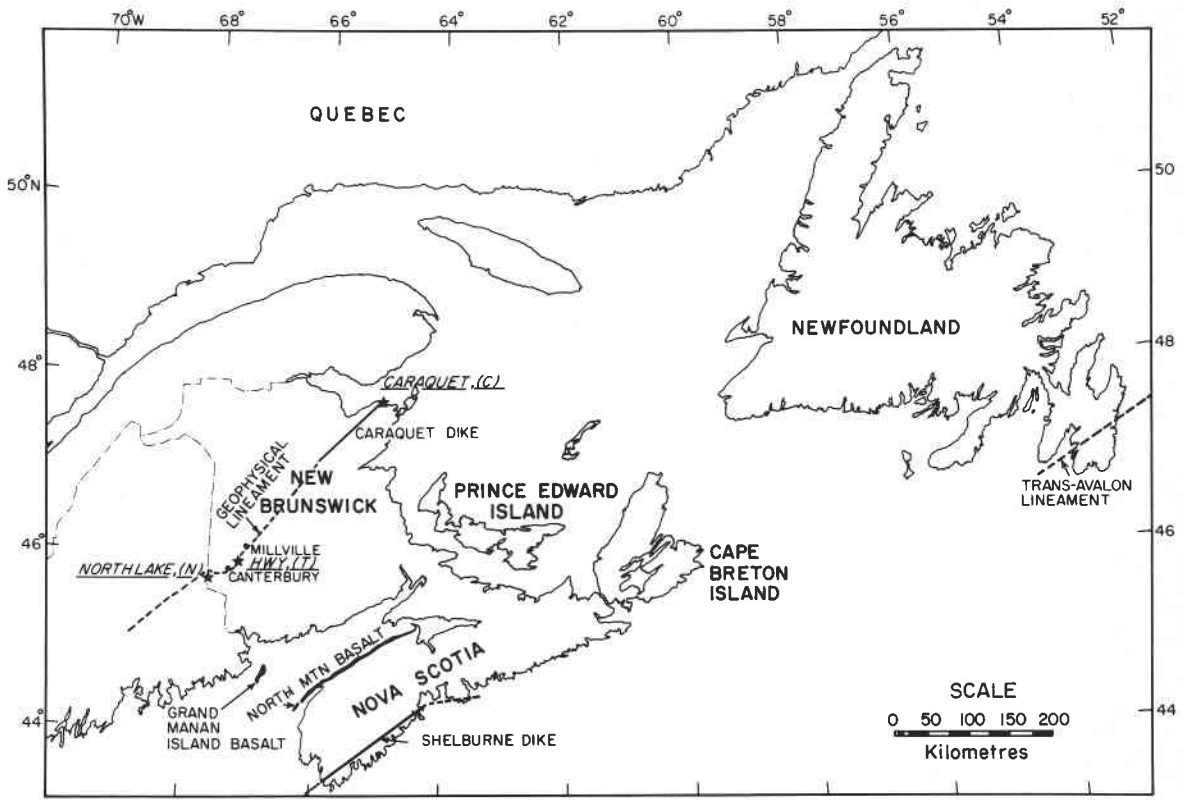


FIG. 1. Location of paleomagnetic sampling sites C, T, and N (marked by star symbol) along the trans-New Brunswick aeromagnetic lineament (Burke *et al.* 1973). The position of the Caraquet diabase dike is identified along the geophysical lineament by a solid line. The Late Triassic – Early Jurassic igneous activity in other parts of the Canadian Appalachians is also indicated.

and aeromagnetic lineaments; and (6) minor diabase dikes and sills at three widely separated localities along a prominent aeromagnetic lineament in the Avalon Peninsula, Newfoundland (Papezik *et al.* 1975; Hodych and Hayatsu 1980; Papezik and Hodych 1980).

Hodych and Hayatsu (1980) have reported two episodes of igneous intrusion, at 201 and 191 Ma (Late Triassic to possible Early Jurassic) along the trans-Avalon aeromagnetic lineament. They also obtained a date of 201 Ma as an age of emplacement for the Shelburne dike and suggested that it is different from the age of North Mountain Basalt (191 Ma; Hayatsu 1979). Burke *et al.* (1973) assigned a Late Triassic – Early Jurassic age for the Caraquet dike based on K–Ar dates obtained from three samples (167 ± 28, 180 ± 31, and 180 Ma) in the vicinity of Caraquet (Poole 1968 *in* Wanless *et al.* 1972). For the Newark Group volcanics in eastern United States, Armstrong and Besancon (1970) have reviewed K–Ar dates that cluster about 200 Ma, with a scatter toward lower values. The ages of mafic dikes in the New England region span Late Triassic – Early Cretaceous but form two major age groups of 180–200 and 96–136 Ma time intervals (Foland and Faul 1977; McHone 1978). Radiometric age determinations on diabase dikes and sills from northwestern Liberia (Dalrymple *et al.* 1975) and High Atlas (Manspeizer *et al.* 1978) regions also indicate that the age of emplacement of basalts is almost always restricted to the uppermost part of the Triassic. Different models of dike emplacement during the Late Triassic – Early Jurassic were suggested (De Boer 1967, 1968; May 1971; Ballard and Uchupi 1975; De Boer and Snider 1979), based on the systematic pattern of Triassic dikes in the Appalachians (King 1961, 1971).

The age of magnetization of igneous rocks in the central Appalachians is reported in the paleomagnetic literature (Table 1). Paleomagnetic studies have also been reported for areas in the Canadian Appalachians. These include the North Mountain Basalt, the Grand Manan Island Basalt, the Shelburne dike, and the Avalon sills and dike (Table 1). In this paper, we report a paleomagnetic study of samples of the Caraquet dike

TABLE 1. Paleomagnetic pole positions from Late Triassic – Early Jurassic igneous rocks of the northern and central Appalachians

Location	Sites/ samples	Pole position		δp (°)	δm (°)	α_{95} (°)	Reference
		Long. (°E)	Lat. (°N)				
Newfoundland							
1. Avalon lineament sills and dike	2/12*	88	73	3	4		Hodych and Hayatsu (1980)
Nova Scotia							
2. Shelburne dike	2/11*	98	69	3.5	5.5		Larochelle and Wanless (1966)
3. North Mountain Basalt	17*/28	113	66	6	10.5		Larochelle (1967)
4. North Mountain Basalt "lower" samples	—/17*	67	67	4.5	6.5		Carmichael and Palmer (1968)
5. North Mountain Basalt "upper" samples	—/21*	149	71	4	5.5		Carmichael and Palmer (1968)
New Brunswick							
6. Grand Manan lavas	—/8*	100.5	80.5	17	23.5		Carmichael and Palmer (1968)
7. Caraquet dike							
a. Caraquet	4*/20	143	78	9	12		This paper
b. North Lake	2/6*	133	74	7	11		This paper
c. Trans-Canada Highway	2/10*	82	62	5	8		This paper
Massachusetts							
8. Granby, Holyoke lavas (Meriden Formation)	5*/16	88	59	6	11		Irving and Banks (1961)
Connecticut and Maryland							
9. Diabase dikes and sills	61*/—	101	69			2	Smith (1976)
10. West Rock and Mt. Carmel sills	8*/41	88	62			4	De Boer (1968)
Pennsylvania							
11. Diabase intrusions	78*/—	104.5	62			1.5	Beck (1972)
New Jersey							
12. Newark lavas	5*/69	113	64			9	Opdyke (1961)

NOTES: δp, δm = semi-axes of ovals of 95% confidence; α_{95}° = semi-angle of cone of 95% confidence.
*Pole position based on site- or sample-mean remanence direction.

system in New Brunswick. This study is based on samples from three widely separated localities where outcrops of only limited exposure are available.

Geologic setting and paleomagnetic sampling

The dikes of Late Triassic – Early Jurassic age can be traced by outcrops and ground and airborne magnetic anomalies, which produce a geophysical lineament that extends from Caraquet in northeastern New Brunswick to eastern Maine. Although the amplitude of vertical intensity over the dike varies from 400 to 1200 nT, the aeromagnetic expression is very weak (Hamilton and Gupta 1972). The dikes are exposed only at a few localities where they vary in width from 5 to 60 m. The rocks are typically dark grey to black, massive and dense and mostly fine grained with no evidence of metamorphism. The average strike is northeasterly with dips vertical to subvertical. The dike pattern is generally discontinuous but locally offset and forms up to three or more closely spaced dikes, perhaps representing separate injections (Hamilton and Gupta 1972; Burke *et al.* 1973).

For the present study, oriented hand samples were obtained (Fig. 1) from: (1) near Caraquet (C; 47°47′45″N, 64°56′15″W), where the dikes intrude flat-lying Pennsylvanian sediments; (2) a road cut on the Trans-Canada Highway (T; 45°53′09″N, 67°26′04″W) about 1 km west-southwest of Highway 122 (map area H22, Venugopal 1979) where the dike intrudes light greyish quartzite and siltstone of Cambro-Ordovician age; (3) in the locality of North Lake (N; 45°46′52″N, 67°45′20″W) where the dike intrudes the Devonian granite locally known as the Pokiok pluton (map area G23, Lutes 1979). From locality C, 20 samples were collected at four sites from three or four closely spaced dikes. From locality T, the collection comprised five samples from two sites. From locality N, three samples from two sites were obtained. The width of the sampled dikes varied from 5 to 20 m. In view of the limited nature of this study (based only on three localities) and because of the lack of reliable radiometric data from the sampling sites, the present investigation is no more than a first step to find out whether or not the sampled material belongs to a single dike system.

Magnetization measurements

The direction (D = declination, I = inclination) and magnetic moment, M, of natural remanent magnetization (NRM) were measured with a Schonstedt digital spinner magnetometer (model DSM-1). Room-temperature magnetic susceptibilities, k, were measured in the Earth's field ($H = 0.05$ mT) with a Cenco MS-3B unit. Step-wise demagnetization tests were carried out using: (1) an alternating field (AF) demagnetizer with a three-axis tumbler in about 5 nT residual fields (Seguin 1975); and (2) a Schonstedt thermal demagnetizer (TSD-1) in residual fields of less than 15 nT. For the NRM of each oriented sample a minimum of two to a maximum of four specimens (2.5 cm in diameter; 2.2 cm in height) were used to obtain a sample-mean direction. Along with the optical examination (thin and polished sections), the diabase samples were subjected to isothermal remanent magnetization (IRM) up to about 1 T using a direct current (DC) electromagnet to determine the magnetic mineralogy. In addition measurements of (1) thermal variation of susceptibility in the temperature range 20–700°C with a low field (0.031 mT) alternating current (AC) susceptibility bridge (Pätzold 1972), and (2) hysteresis properties in magnetic fields of $\simeq$120 mT, with a hysteresis loop-tracer (Likhite *et al.* 1965), were made on three diabase specimens.

Paleomagnetic study

NRM and induced magnetization

Based on sample means, the NRM site directions from the three localities (C, N, and T) indicate the varying influence of the present Earth's field (PEF) (D = 338°, I = +73°). The C and N site-mean NRM values spread from a direction not far from the PEF towards the T site-mean directions (Fig. 2, Table 2).

The sample-mean values of NRM intensity, J_n, and susceptibility, k, indicate that they range from 6×10^{-1} to 2.5 A m^{-1} in remanence but within a relatively narrow range in induced magnetization, kH, $3–9 \times 10^{-1}$ A m^{-1}, with Koenigsberger ratios ($Q_n = J_n/kH$ in H =

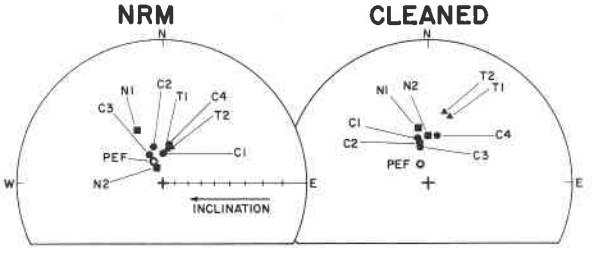


FIG. 2. Site-mean directions of magnetization plotted on a Wulff net for NRM and after AF and thermal (combined) cleaning. C, N, and T sampling sites are identified, respectively, with solid circles, squares, and triangles. The solid symbols denote downward (positive) inclinations. The star represents the present Earth's field (PEF) at the sampling sites.

TABLE 2. Site-mean directions of magnetization for New Brunswick dike

	NRM				AF(15–50 mT)				Thermal (300–550°C)*				Combined			
Site	N	(n)	D (°)	I (°)	K	N	(n)	D (°)	I (°)	K	N	(n)	D (°)	I (°)	K	D (°)	I (°)	K	α_{95} (°)
C1	5	(15)	1	+66	22	5	(10)	343	+54	59	5	(5)	350	+54	34	347	+54	47	7.1
C2	5	(13)	346	+61	108	5	(8)	348	+58	190	3	(5)	343	+57	334	347	+58	238	3.6
C3	5	(17)	335	+66	49	5	(10)	353	+59	79	4	(7)	340	+61	131	347	+60	89	5.5
C4	5	(16)	8	+60	10	4	(10)	8	+51	39	3	(6)	12	+53	353	10	+52	69	7.3
Mean	4		353	+64	118	4		354	+56	128	4		352	+57	84	353	+56	120	8.4
N1	3	(10)	335	+44	10	3	(6)	352	+48	64	3	(4)	345	+47	63	349	+46	73	7.9
N2	3	(9)	337	+75	47	3	(5)	346	+54	310	1	(4)	341	+48	—	359	+53	80	10.4
Mean	2		335	+60	—	2		349	+51	—	2		343	+47	—	353	+50	—	—
T1	5	(16)	6	+64	6	5	(10)	15	+36	69	5	(6)	19	+40	43	17	+38	57	6.5
T2	5	(17)	9	+62	28	5	(12)	13	+33	80	3	(5)	9	+39	224	12	+35	32	10.0
Mean	2		8	+63	—	2		14	+35	—	2		14	+39	—	14	+37	—	—

NOTES: N = number of samples from n specimens; D, I = mean declination and inclination of site; K = estimate of Fisher's precision parameter; α_{95} = radius of cone of confidence at 95% probability level.
*Ranges of T_{UB} over which mean directions are calculated for C, N, and T sites are 300–500°C and 500–550°C.

0.05 mT) between 1.01 and 7.24. According to De Boer and Snider (1979), the Late Triassic – Early Jurassic diabases were part of a single intrusive system. They found that roughly 80% of the diabase samples have Q_n values between 0.4 and 10 (Fig. 3) with a median value of 1.1 and that the induced magnetization is predominantly responsible for the magnetic anomaly pattern observed over dikes in the Appalachians. Interestingly enough, this is in striking contrast to the younger Deep Sea Drilling Project (DSDP) basalts, which are characterized by Q_n values with a geometric mean of about 10 (Lowrie 1977).

The initial magnetic properties, J_n and k, of the Caraquet dike compare very well and fall within the range of values for the other dikes in the Appalachians (Table 1 of De Boer and Snider 1979). A comparison of these properties from the other rock units (North Mountain Basalt, Carmichael and Palmer 1968; Avalon sills and dikes, Shelburne dike, Hodych and Hayatsu 1980; diabase sheets, Beck 1966) reveals that the range in values of J_n and kH are not significantly different from each other despite local variations within each of the diabase units caused by factors such as variable width of dikes, cooling rates of respective intrusions, or magmatic differentiation undergone by diabase sheets, etc. (Fig. 3).

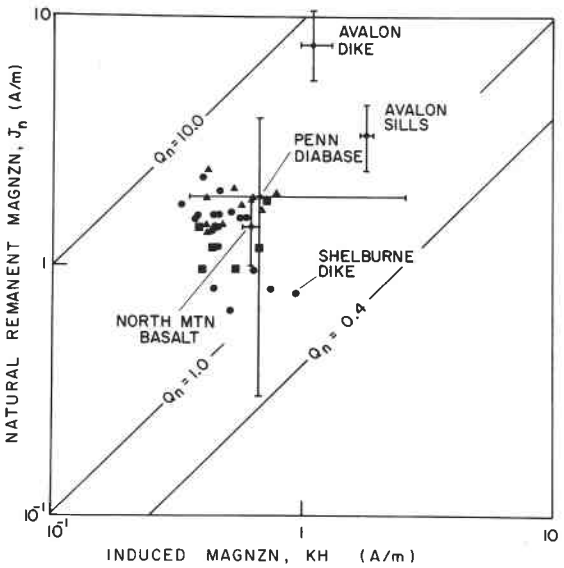


FIG. 3. Individual arithmetic sample-mean values of NRM intensity (J_n in amperes per metre) are plotted against the corresponding values of induced magnetization (kH in amperes per metre) for samples from sites C (circles), N (squares), and T (triangles). The resulting slopes indicate the value of the Koenigsberger ratio, Q_n ($=J_n/kH$). For comparison, published data from other contemporaneous igneous rocks are shown. These include: (1) Hodych and Hayatsu (1980) from three sills (one site) and one dike (one site) along the Avalon magnetic lineament. Also a mean of two samples taken from the Shelburne dike (Fig. 1). (2) Carmichael and Palmer (1968) from 40 samples of North Mountain Basalt. (3) Beck (1966) on 219 specimens from six differentiated Pennsylvania diabase sheets. The vertical and horizontal bars represent the range of standard deviations from the mean in J_n and kH, respectively. The values of Q_n between 0.4 and 10 represent the range computed for Late Triassic – Early Jurassic diabase dikes from the Appalachians (De Boer and Snider 1979).

AF and thermal demagnetization

One or two specimens from each of the 36 samples were subjected to AF demagnetization in steps of 5 mT in the range 5–50 mT and in steps of 10 mT in the range 50–80 mT. The demagnetization results were subsequently analysed by using Zijderveld orthogonal plots (Zijderveld 1967). Although there is no significant increase in the precision, K, of the mean direction after AF demagnetization there is a shift of mean direction away from PEF at all eight sites, indicating removal or partial removal of minor viscous remanence. For both within and between C and N localities, the agreement in directions was much improved after the removal of the viscous component in a field of 5–15 mT. A northwesterly stable remanence with steep downward inclinations (D = 353°, I = +56°) characterized by a linear vector trajectory is isolated in the 15–40 mT range in the majority of C and N specimens. They lost 50% of their NRM intensity by 10–30 mT and more than 90% by 80 mT. In contrast, the orthogonal plots for T specimens show a northeasterly directed magnetization with relatively shallow inclinations (D = 14°, I = +35°) removed between fields of 10–50 mT. Two representative demagnetization trajectories for specimens C153 (typical of C and N specimens) and TB1 (typical of T specimens) are shown in Fig. 4. The corresponding coercivity spectra derived from the respective normalized decay curves (Fig. 4) indicate that the coercivities are well below 40 mT. No characteristic coercivity values were found to be associated with the northwesterly or northeasterly directed remanences. This overlap in the coercivity values for both the remanences is a common feature in the majority of the samples.

About 42 companion specimens were thermally demagnetized at 100°C intervals between 100 and 500°C and at temperatures of 525, 540, 565, 580, and 600°C, respectively. Thermal treatment by 100°C could remove low-stability viscous remanence in the specimens and the isolation of the stable remanence direction at temperature steps between 300 and 550°C. This direction is similar to the one observed between 15 and 40 mT during AF treatment (Table 2). After treatment at 550°C and higher, the directions become random and no thermally stable remanence appears to survive. Although the directional behaviour of the samples in both treatments is almost similar their common decay characteris-

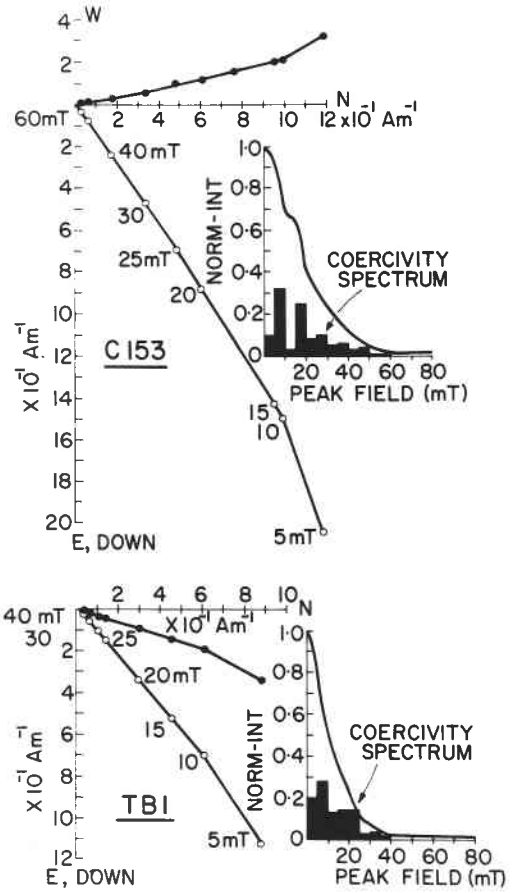


FIG. 4. Representative orthogonal AF demagnetization diagrams (Zijderveld 1967) of successive end points of magnetization vector for two specimens as indicated. Solid (open) circles represent projections on horizontal (vertical) plane. Numbers next to open circles represent the demagnetization steps in milliteslas. Units indicated for axes are in 10^{-1} A m^{-1} ($=10^{-4}$ emu cm^{-3}). Corresponding normalized intensity (J/J_n) curves and coercivity spectra are shown to the right of the orthogonal plots. The spectra are calculated using the gradient of the demagnetization curves between successive 10 mT steps. Note that the coercivity values are well below 40 mT and the two spectra overlap with each other.

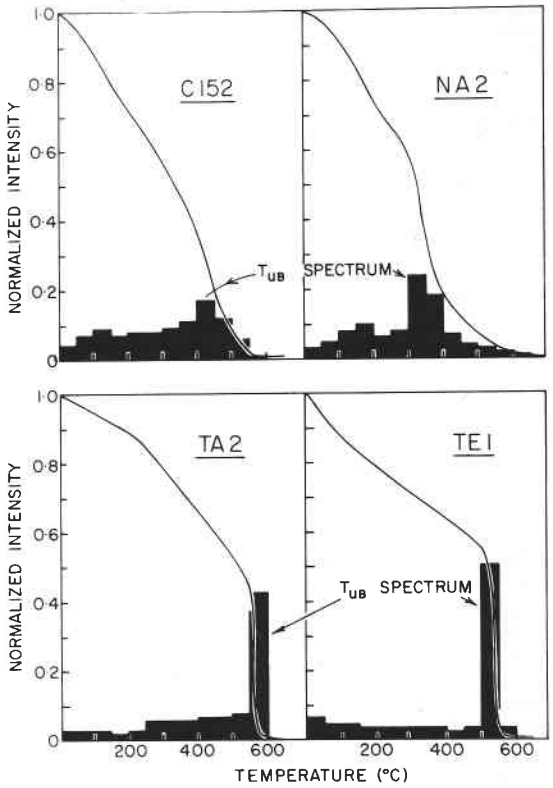


FIG. 5. Representative normalized intensity (J/J_n) decay curves and the corresponding unblocking temperature (T_{UB}) spectra after step-wise thermal demagnetization for four specimens as indicated. The T_{UB} spectra are calculated using the gradient of the demagnetization curves between successive 50°C steps. Note that the spectra (top) are distributive and below 500°C; spectra (bottom) are discrete as shown by peaks just below the corresponding Curie points (Fig. 7*b*).

tics of normalized remanence were found to differ (Fig. 5). The AF decay curves tend to fall steeply to one half their initial value by 15–30 mT (Fig. 4) while the thermal decay curves reach one third of their initial value around 500°C. In contrast to the commonly observed overlap in the coercivity spectra, the unblocking temperature (T_{UB}) spectra derived from the thermal decay curves show a striking difference between C, N, and T specimens (Fig. 5). Whereas the thermal curves for C and N specimens indicate the decay of NRM over a wide range of T_{UB} (100–400°C), the decay curves for T specimens show the loss of NRM over a narrow temperature interval (500–550°C) just below the Curie point. Following the classification of blocking temperature spectrum by Irving and Opdyke (1965), the C and N samples carry a thermally "distributed" component and the T samples possess a thermally "discrete" component. But the coercivity and T_{UB} spectra with their corresponding characteristics of normalized remanence (Figs. 4, 5) suggest magnetite and (or) titanomagnetite to be the common carrier of remanence for both cases.

Magnetic mineralogy

Three diabase samples from each of the three sampling regions exhibit ophitic to subophitic texture with a uniform content of plagioclase feldspar (40–55%), augite (40–50%), plus minor amounts of olivine (2–5%) and opaque minerals (1–2%). The dominant opaque mineral is probably magnetite, which occurs in skeletal form. From optical examinations of the corresponding polished sections under microscope, the

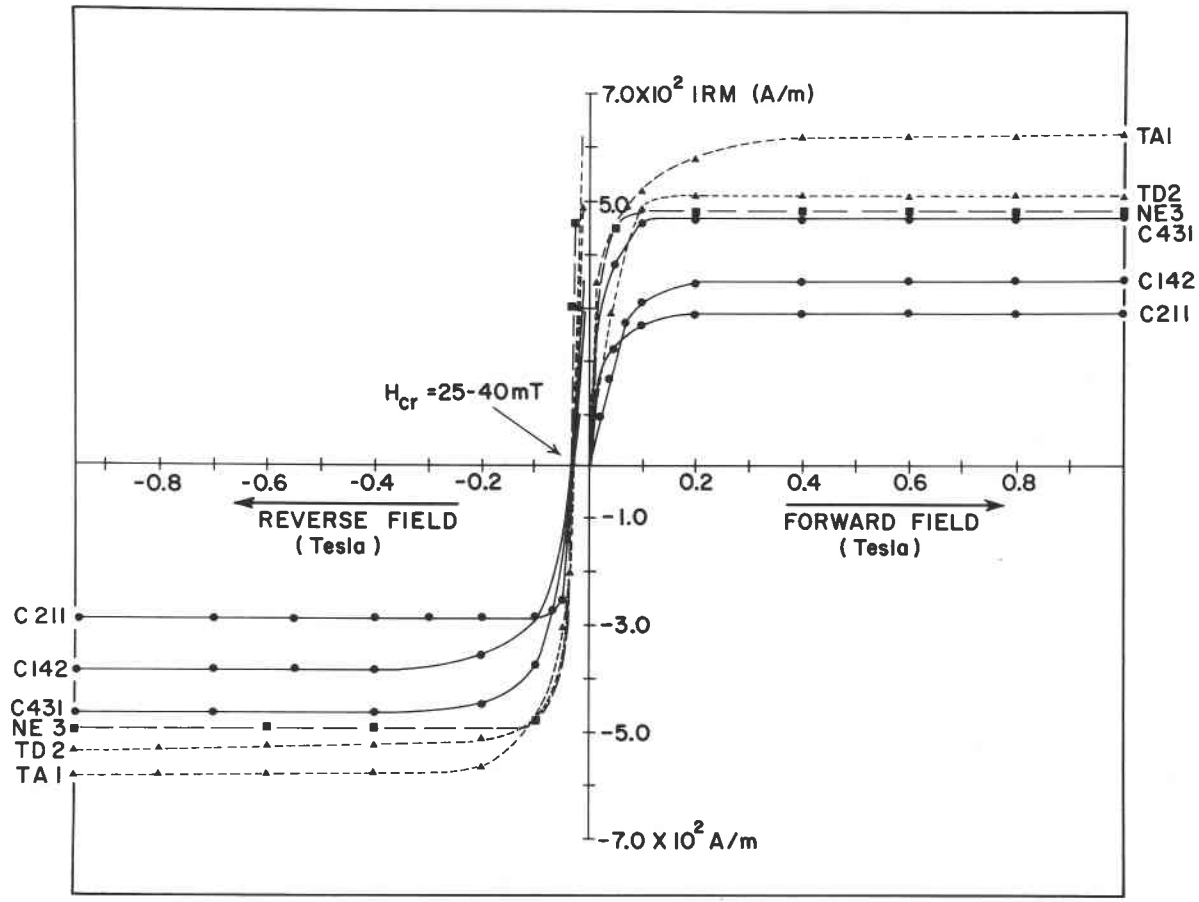


FIG. 6. Acquisition of isothermal remanent magnetization (IRM in amperes per metre) for six specimens (as indicated) at room temperature, in forward and reverse fields up to 1 T. The plotted points represent values measured in a zero field after magnetization in the fields shown. The intersection of curves in the reverse field axis corresponds to the values of coercivity of remanence (H_{cr}) as indicated. The other symbols are as in Fig. 3.

opaques are identified as titanomagnetite grains with exsolution textures of ilmenite lamellae along [111] planes. The estimated grain sizes of titanomagnetite vary from 10 to 30 μm. Based on the Watkins–Haggerty oxidation scale (Watkins and Haggerty 1967), the diabase samples fall under stages I and II, indicating a low oxidation state. Examination of hand samples and optical studies of thin and polished sections reveal no significant differences between C, N, and T samples in terms of either composition or opaques.

IRM experiments based on six diabase specimens in applied fields up to 1 T reveal that their acquisition curves reached saturation around 150 mT, after a very rapid initial rise, despite some differences in the magnitude of saturation IRM. The values of remanent coercive force, H_{cr}, vary in a very narrow range, i.e., 17–37 mT (Fig. 6). These IRM characteristics show that the main magnetic mineral is probably either magnetite or titanomagnetite. Also, lack of noticeable inflections in the reverse field IRM curves suggests the presence of only a single mineral responsible for the remanence. The values of H_{cr} = 17–37 mT obtained in the IRM experiments agree well with the value of median destructive field $\bar{H}_{1/2}$ = 15–30 mT for the majority of samples. This observation is in accordance with the findings of Dunlop and Bina (1977) that $H_{cr} \simeq H_{1/2}$.

Thermal variation of magnetic susceptibility in the temperature range 20–700°C revealed Curie points, T_c, close to 525–575°C, indicating the presence of magnetite (Fig. 7*b*) and further confirming the characteristics of IRM acquisition curves. High-field (120 mT) hysteresis curves at room temperature (not shown) yielded mean values of coercivity H_c = 20 mT and J_r/J_{max} = 0.2; the corresponding mean value computed for H_{cr}/H_c = 1.1. The range of these hysteresis ratios corresponds to titanomagnetites in single domain to pseudo-single domain range (Day *et al.* 1977). The mean magnetization directions (D = 353°, I = +53° for C and N sites; D

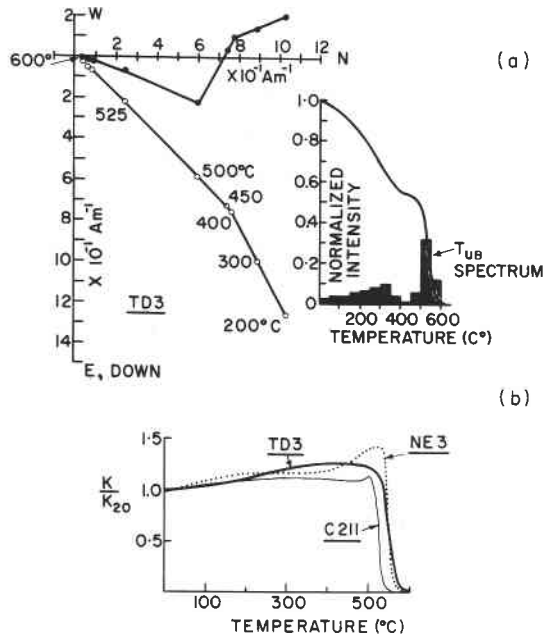


FIG. 7. (*a*) Orthogonal thermal demagnetization diagram (Zijderveld 1967) of successive end points of magnetization vector for one specimen as indicated. Corresponding thermal decay curve and the T_{UB} spectrum are shown at the right. Note the distinctly separate wide (100–400°C) and narrow (500–600°C) range in T_{UB} values. The other symbols are as in Fig. 4. (*b*) Normalized susceptibility (k) versus temperature (T in degrees Celsius) for three specimens as indicated in 0.031 mT using the susceptibility bridge (Pätzold 1972).

$= 14°$, $I = +37°$ for T sites, Table 2) characterized by linear vector trajectories in AF (15–50 mT) and thermal (300–500°C for C and N sites; 500–550°C for T sites) treatments could then be attributed to the presence of titanomagnetite in the diabase samples.

Paleomagnetic poles

Site-mean directions were calculated from NRM, AF (15–40 mT), and thermal (300–550°C) values. Since the mean directions obtained from AF and thermal treatments agree with each other the results were combined for each sampling locality (Fig. 2, Table 2). The combined AF and thermal results yield a higher K value (two to nine times) relative to its NRM value for each site. Contact tests to study the age and stability of magnetization at each sampling locality could not be performed due to the weak ($\leq 10^{-3}$ A m^{-1}) and unstable nature of remanence of the older sediments (C, T) and granites (N) through which the sampled dikes were intruded. Paleomagnetic poles corresponding to the locality-mean directions were calculated and listed together with the published Late Triassic – Early Jurassic poles derived from other igneous rock units from the Appalachians (Table 1, Fig. 8). The poles obtained in

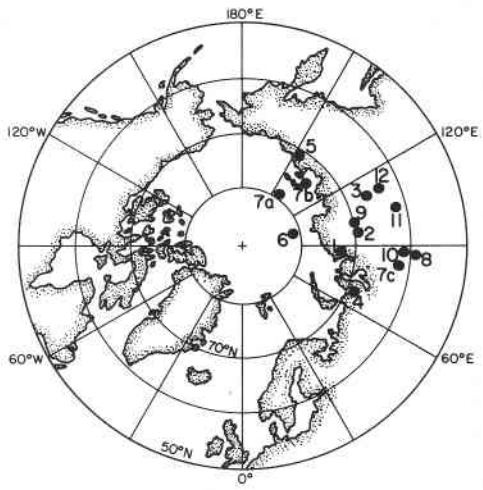


FIG. 8. Late Triassic – Early Jurassic pole positions (Table 1) plotted on polar projection of the northern hemisphere north of latitude 50°N.

the present study, based on a limited collection, are considered as virtual paleopoles instead of paleomagnetic poles since they represent only spot readings of the paleomagnetic field (McElhinny 1973, p. 26).

Consideration of Table 1 and Fig. 8 suggests some common features: (1) the majority of poles obtained from the Canadian Appalachians probably represent virtual poles since the cooling times of the rock units involved are shorter than 10^4 years, which may be necessary to average out secular variation; (2) these poles, 1–7, are derived from a data set of inadequate paleomagnetic sampling, perhaps because of poor exposures of these igneous rocks in contrast to the data from well exposed igneous units in the central Appalachians; (3) no reversals of the field are recorded in the Upper Triassic igneous rocks; (4) active drifting of the North American continent during the Late Triassic amounts to perhaps several tens of degrees (Beck 1972); (5) these studies also indicate a tendency for the measured NRM directions to fall into two groups: group A with north or northeasterly declinations and group B with north or northwesterly declinations. The group A directions commonly possess relatively shallow downward inclinations compared with those of group B, which are predominant with steep downward inclinations. The group B directions also show an oval distribution with a spread not far from the PEF towards group A, while the demagnetization of group B results in a shift of the mean direction away from the PEF (which may mean removal of viscous remanence), noticeably different from the mean direction obtained for group A after treatment.

For example, poles 4 and 5 (Table 1) are derived from two lava flows that are stratigraphically separated; their corresponding magnetizations are proved to be primary,

as evidenced by a conglomerate test. Carmichael and Palmer (1968) found no obvious differences in the magnetic properties of the older (pole 4, group A) and younger (pole 5, group B) flows; the age difference between the times of eruption of lavas is not known but the two groups of directions were attributed to incomplete averaging of secular variation. However, the pole reported in their study, corresponding to remanence obtained from the mean of all the samples (sites) (107°E, 73½°N (104°E, 73°N)), differs from another pole, 3, (113°E, 66°N) derived also from the North Mountain Basalt (Larochelle 1967). Still, the agreement between the poles obtained in the present study and those from the North Mountain Basalt (7a, 7b with 5 and 7C with 4) is significant (Fig. 8). The mean age of magnetization corresponding to paleopoles 8 and 10 from the central Appalachians and virtual poles 1 and 2 from eastern Newfoundland and Nova Scotia are well established as Late Triassic (≈200 Ma). These poles are not different from virtual pole 7C, which suggests that the dike sampled at locality T is at least different in age and perhaps older than the dikes at C and N sampling areas. Radiometric data obtained from only three samples at locality C suggest a mean K–Ar age of 180 Ma (Burke *et al.* 1973), which corresponds to an Early Jurassic age (Van Eysinga 1975). This may then mean that the dikes at localities C and N differ in age from the dike at locality T by at least 20 Ma.

Furthermore, virtual poles 7a and 7b differ significantly from the rest of poles 1–12 (except pole 5) but fall nearer to the younger Mid- to Late Jurassic average paleopoles, while pole 7C agrees with the average Late Triassic pole computed, based on a statistical procedure of moving averages (Irving 1979). However, in the absence of a contact test, it is hard to judge the relative ages of magnetization of dikes at the three localities.

"Younger" magnetizations

In a recent paleomagnetic study on Tasmanian dolerites of Jurassic age from Australia, Schmidt (1976) reports the presence of two components, one Early Jurassic and the other probably Late Cretaceous and related to rifting but with similar magnetic properties. This prompted us to investigate the possible presence of a highly stable secondary remanence in the Late Triassic – Early Jurassic diabases of the Appalachians. An evidence for such likely occurrence of secondary magnetization is found only in three T samples upon thermal treatment. They show the presence of both northwesterly (with a wide range in T_{UB} (100–400°C), similar to many C and N samples) and northeasterly (with a narrow range in T_{UB} (500–600°C), similar to other T samples) remanences associated by linear vector trajectories (specimen TD3, Fig. 7*a*). In contrast to the AF (Fig. 4) and thermal (Fig. 5) spectra of the majority of samples, the low (100–400°C) and high (500–600°C) T_{UB} spectra in the specimen TD3 do not overlap with each other. Based on this limited evidence, the acquisition of northwesterly magnetization observed in C and N sites could be attributed to a build-up of viscous partial thermoremanence (VPTRM) imposed at moderate temperatures. The northeasterly remanence ($D = 15°$, $I = +35°$) associated with a narrow range in T_{UB} just below the Curie point (Fig. 5, Fig. 7*b*) could be the magnetization acquired at the time of emplacement (≈200 Ma). Interpretation of "younger" magnetizations (such as northwesterly remanence $D = 350°$, $I = +53°$ seen in C, N sites) as overprints acquired later in geological time and examples of rock formations that show no petrological signs of baking but possess stable secondary components are discussed in paleomagnetic literature (Pullaiah *et al.* 1975).

Examples suggesting the possible presence of secondary directions indicated by variability in AF or thermal decay curves could be inferred from the earlier studies (Fig. 2 of Irving and Banks 1961; Fig. 7 of Opdyke 1961). Similar to thermal decay curves with a wide and narrow range in T_{UB} spectrum obtained in the present study (Fig. 5), thermal demagnetization curves shown for North Mountain Basalt (Fig. 2 of Carmichael and Palmer 1968) also exhibit a wide variation in T_{UB} (Curie points vary from 175 to 530°C). Again the AF decay curves obtained from companion specimens in their study indicate overlap in the coercivity values just as in the present study (Fig. 4). De Boer and Snider (1979) cite local differences in the inclinations (+11 to +54°) of thermoremanent magnetization directions obtained from contemporaneous North Carolina dikes with a clear tendency towards higher inclination values with decreasing age. They attributed the younger magnetizations to a "chemical remanent magnetization(?)" possibly associated with postemplacement low-grade regional metamorphism (Fig. 4*B* of De Boer and Snider 1979). In summary, these studies indirectly suggest the possible presence of younger magnetizations in Late Triassic – Early Jurassic rocks, which we believe could be overprints probably associated with rifting and opening of the Atlantic in Middle Jurassic time. Based on the predicted thermal history at the time of rifting (Keen 1979) and magnetization changes caused by tectonic events such as burial and uplift (Pullaiah *et al.* 1975), we tend to believe that younger magnetizations (as shown by C, N sites) could be related to VPTRM originating at the time of opening of the Atlantic.

But the present study is based on only three limited diabase exposures and the evidence is not strong enough to speculate on the possibility of an overprint. The most striking characteristic of the Caraquet dike is its discontinuous nature, offset locally by faults. At least five breaks in continuity occur between Caraquet Island

and Wayerton (Fig. 2 of Burke *et al.* 1973) and it is not known with certainty if this constitutes one dike system representing a single intrusive event. Differences in the mean remanence directions between sampling sites T ($D = 14°$, $I = +37°$) and C, N ($D = 353°$, $I = +53°$) and geophysically inferred differential offset in the dike system may indicate different periods of dike emplacement with a local age variation ranging, respectively, from Late Triassic (200 Ma) to Middle Jurassic (180 Ma), a maximum difference of 20 Ma. Detailed paleomagnetic studies are in progress on diabase outcrops recently found along the geophysical lineament in New Brunswick and Maine (1) to test if the age differentials speculated here hold true for other parts of the dike system and (2) to determine whether the observed stable remanence directions represent different periods of dike emplacement or the effect of moderate temperature VRM associated at the time of rifting in the Middle Jurassic.

Conclusions

Paleomagnetic study of dike samples taken from parts of the northeasterly trending Caraquet dike system, associated with the trans-New Brunswick aeromagnetic lineament, resulted in the determination of paleopoles 7a (143°E, 78°N), 7b (133°E, 74°N), and 7c (82°E, 62°N) for the three widely separated sampling areas C (353°, +56°), N (353°, +50°), and T (14°, +37°), respectively. Determination of these virtual pole positions is based on stable remanence directions carried by titanomagnetite. In contrast to pole 7c, which agrees well with other Late Triassic – Early Jurassic poles for the Appalachians, poles 7a and 7b are significantly different and fall in the general area of average paleopoles computed for Middle–Late Jurassic age. A likely possibility that the age of the dike at T differs from that of dikes at C and N by at least 20 Ma is suggested but this is tentative since paleomagnetic data are based on a limited number of samples. In the absence of reliable radiometric data, it is premature at the present stage to predict with certainty the likely occurrence of younger magnetization in some of these dikes.

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